Alex Vedernikov

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Summary

Senior Data Scientist, 11 years of experience, PhD in Computer Science. Geospatial data science at GapMaps and Aurecon: spatio-temporal analytics on device-location data, automated geospatial pipelines and deep learning on satellite imagery. Built a pricing system adding ~600K EUR of revenue per year relative to standard prices.

Skills

Geospatial & GIS: Sentinel satellite imagery, GeoPandas, GDAL, QGIS, ArcGIS, PostGIS, Apache Sedona, Google Maps API, CARTO, Kepler.gl, OpenStreetMap
Machine learning & statistics: deep learning on satellite imagery, classification, regression, forecasting, optimisation, XGBoost, A/B testing
Tools: Python, SQL, PostgreSQL, Snowflake, FastAPI, PyTorch, Docker
Cloud: AWS, GCP, Azure, Databricks
Generative AI: LangGraph, LangChain, RAG, agentic AI, Claude Code
Dashboards: Looker Studio, Dash, Streamlit, Shiny, Tableau, Power BI

Experience

DayTrip, Senior Data Scientist — Prague (remote), Nov 2023 – May 2026

- Built a fully automated pricing system that dynamically adjusted customer prices on the website and driver prices in the app, adding ~600K EUR of revenue per year relative to standard prices.
- Analysed driver distances using the Google Maps API as part of load optimisation within the pricing system.
- Unified data across spatio-temporal dimensions such as country, route and time to departure.
- Designed the pricing data architecture, from raw data (Google Analytics sessions, partner API requests, website logs) to dashboard-ready layers, used by several teams.

GapMaps, Senior Data Scientist — Melbourne (hybrid), Mar 2022 – Oct 2023

- Set up a pipeline ingesting ~100M device locations per day and extracted spatio-temporal visitation patterns (including home and office locations) for major retail areas and shopping malls in nine countries.
- Developed a fully automated pipeline producing time-sensitive street-traffic metrics from device-generated data across major urban centres in nine countries, replacing expensive third-party solutions.
- Conducted an analysis of swimming-pool detection from imagery and incorporated the results into the framework for identifying affluent areas, estimating consumer classes in combination with demographic data.
- Built a country-specific retail hierarchy classifying small, medium and large shopping malls from multi-source data, supporting client network strategy and new-location selection across five countries.
- Responsible for purchasing mobile location data from multiple providers, including assessment of data bias and suitability for client needs.
- Delivered location-intelligence analytics for clients across fast food, fitness, childcare and retail.

Aurecon, Data Scientist — Melbourne, Nov 2019 – Mar 2022

- Detected road kerbs on satellite imagery using deep learning and compared the detections with cadastral data, integrating both sources to update cadastral GIS data for Wellington City Council.
- Used ~500M user mobile locations to find spatio-temporal visitation patterns at Sydney beaches for Sydney Water.
- Conducted multivariate customer segmentation and spatial analysis of customer behaviour for a government transport authority.
- Constructed a data pipeline for a port scheduling optimisation project (Gladstone Ports Corporation).
- Supervised two interns on data science projects.

Education

PhD **University of Melbourne, Australia**, Computer Science — 2014 – 2019
- Thesis: Optimal route planning for hitchhiking

MSc **National Chiao Tung University, Taiwan**, Computer Science (GPA 4.00) — 2012 – 2014

BSc **Taras Shevchenko National University of Kyiv, Ukraine**, Computer Science (GPA 3.97) — 2007 – 2011

Certifications

DeepLearning.AI: AI Agents in LangGraph; Long-Term Agentic Memory with LangGraph (10+ other courses)
Microsoft Certified: Azure Data Fundamentals (2021); Google Cloud Platform Fundamentals: Core Infrastructure and How Google Does Machine Learning (Coursera)

Publications

Four peer-reviewed publications on route planning and location-based data (incl. IEEE MDM 2017, Journal of Location Based Services 2019). Full list: ResearchGate

Additional

Citizenship: Australia, Ukraine
Relocation: Relocation possible
Languages: English (fluent), Russian and Ukrainian (native), Mandarin (beginner)